

PHYS-E0510 – Advanced Computational Methods in Physics D

Session 3: Exact Diagonalization in the Fermionic Basis

Companion notebook: `quantum_interacting_fermions.ipynb` **Prerequisites:** Session 1 (second quantization, the Hubbard model, mean-field theory); Session 2 (exact diagonalization, though we rebuild everything we need directly in the fermionic language here)

Contents and goals

Session 1 built the Hubbard model from second quantization and solved it at the mean-field level, and it ended on a warning: mean-field theory can fail badly, even on the smallest possible example, the Hubbard dimer at half filling. Session 2 picked up right at that failure point and solved the resulting Heisenberg spin model exactly, with no product-state restriction at all, by diagonalizing directly in the spin basis. In this session we go back one step further and diagonalize the *original* fermionic Hamiltonian exactly, working in the fermionic Fock space itself rather than first mapping onto spins. This is not just a change of language: it lets us see things a spin model simply cannot represent, charge physics, a real chemical potential, a genuine insulator-versus-metal distinction, and, most strikingly, a completely numerically exact test of Session 1's mean-field magnetization prediction. We will find that mean-field theory gets the Hubbard chain's magnetization qualitatively wrong, not just quantitatively wrong, which is exactly the kind of failure that motivates doing things exactly whenever we can afford to. We close by watching spin and charge decouple from each other as the interaction grows, a genuinely many-body effect called spin-charge separation. Session 5 will pick this exact fermionic program back up and push it to far larger system sizes using tensor networks.

Learning outcomes

By the end of this session, you should be able to:

- Diagonalize interacting fermionic Hamiltonians directly in Fock space and explain how the Hilbert-space dimension scales with system size and spin.
- Use ground-state degeneracy, a chemical-potential sweep, and the charging energy to identify a charge-density-wave transition and control the ground-state particle number.
- Test Session 1's mean-field magnetization prediction against the exact Hubbard result and explain the disagreement using Lieb's theorem.
- Diagnose spin-charge separation from static and dynamical correlators, and connect the strong-coupling limit to the Heisenberg model of Session 2.

1 From spins back to fermions: the many-body Fock space

Let us start by rebuilding the same many-body machinery we used for spins in Session 2, but now directly in terms of fermionic creation and annihilation operators. Recall from Session 1 that a single-particle Hamiltonian acting on L orbitals is just an $L \times L$ matrix, and diagonalizing it costs polynomial time. The moment we add any quartic (interaction) term, that is no longer true: we

have to diagonalize $\mathcal{H}$ in the full many-body Fock space, spanned by all the ways of distributing electrons among the L orbitals, and that space grows exponentially in L .

Let us count this explicitly, the same way the lecture does. Take two spinless sites, labeled 0 and 1. The many-body basis has exactly four states,

$$|\Omega\rangle, \quad c_0^\dagger |\Omega\rangle, \quad c_1^\dagger |\Omega\rangle, \quad c_0^\dagger c_1^\dagger |\Omega\rangle, \quad (1)$$

the empty state, the two singly-occupied states, and the doubly-occupied state, so $\mathcal{H}$ is a 4×4 matrix in this basis. Add a third site and the counting becomes $1 + 3 + 3 + 1 = 8$ states (one empty configuration, three with one electron, three with two electrons, one fully filled), which is 2^3 . In general, for L spinless orbitals the Fock space has dimension 2^L : every orbital is independently either empty or filled, so we get one binary choice per orbital. This is worth contrasting with the single-particle sector alone, spanned by $c_i^\dagger |\Omega\rangle$ for $i = 1, \dots, L$, which only has dimension L : restricting to one electron is precisely what makes single-particle physics polynomially cheap, and dropping that restriction is precisely what makes many-body physics exponentially expensive.

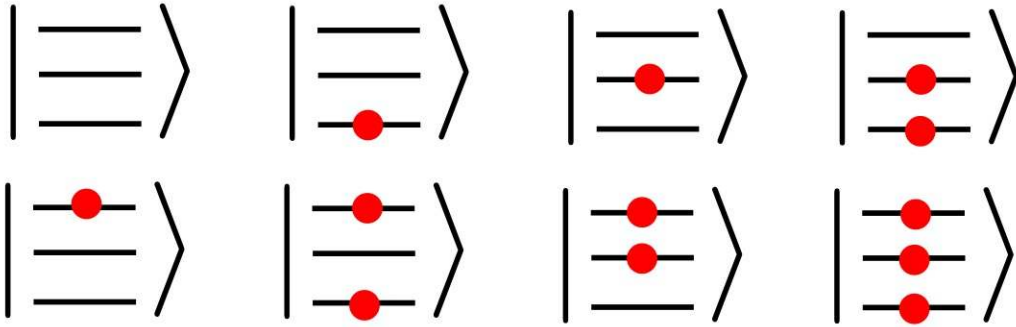


Figure 1: The complete many-body basis of three spinless orbitals, drawn as three levels per state with a dot on every occupied one. Reading across, there is one empty configuration, three with a single electron, three with two electrons and one completely filled state, for $1 + 3 + 3 + 1 = 8 = 2^3$ basis states in total. Every additional orbital doubles this count, because each orbital is independently either empty or filled, and that doubling is exactly the exponential wall that separates many-body from single-particle physics.

If the fermions carry spin, each site has *four* local states, $|0\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle$, exactly as in Session 1. Equivalently, we can think of a spinful chain as $2L$ independent spinless orbitals (spin becomes just another orbital index), so the Fock space has dimension $2^{2L} = 4^L$: twice the exponent of the spinless case, since every site now contributes two binary choices instead of one. It is worth pausing on how this lines up with the spin chains of Session 2: an L -site spin- $\frac{1}{2}$ chain also has a 2^L -dimensional Hilbert space, exactly matching the spinless-fermion count. This is not a coincidence: a rigorous mapping (Jordan–Wigner, which we will meet properly in Session 5) sends spinless fermions to spin- $\frac{1}{2}$ degrees of freedom one-for-one. For now, the practical upshot is simply this:

- solving an interacting **spinless** fermionic chain costs exactly as much as solving a quantum **spin** chain of the same length, $\sim 2^L$;
- solving an interacting **spinful** fermionic chain, like the Hubbard model, costs the square of that, $\sim 4^L$, so at fixed computational budget we can only reach about half the system size.

Just as in Session 2, once we have $\mathcal{H}$ written as an explicit $2^L \times 2^L$ (or $4^L \times 4^L$) matrix in this Fock basis, the ground state is simply its lowest eigenvector, a set of 2^L (or 4^L) complex

coefficients multiplying each basis configuration, and any expectation value we want, $\langle n_i \rangle$, $\langle S_i^z \rangle$, correlators, spectra, follows by ordinary linear algebra on that eigenvector. No approximation, no product-state restriction: this is the entire promise of exact diagonalization, now made good on directly in the fermionic language.

✕ Computational companion

Sections “Creating a fermion_chain object,” “Creating a Hamiltonian,” “Computing the ground state energy,” and “Computing expectation values.” Build a small fermionic chain object, write down a many-body Hamiltonian directly in terms of the fermionic operators, and extract its ground state energy and a couple of expectation values. This is the fermionic analogue of what you did with spin operators in Session 2, and it is worth comparing the two interfaces side by side.

2 The spinless-fermion model: a charge-density-wave transition, computed exactly

Let us put this machinery to work on the simplest interacting model we have: the spinless nearest-neighbor chain from Session 1,

$$\mathcal{H}_{\text{CDW}} = -t \sum_i \left(c_i^\dagger c_{i+1} + \text{h.c.} \right) + V \sum_i \left(n_i - \frac{1}{2} \right) \left(n_{i+1} - \frac{1}{2} \right), \quad n_i \equiv c_i^\dagger c_i. \quad (2)$$

This is exactly Session 1’s $\mathcal{H}_{\text{CDW}}$, just with $n_i \rightarrow n_i - \frac{1}{2}$ inside the interaction, a purely bookkeeping shift: expanding it out just adds a constant plus an automatic chemical potential of $-V/2$ per site, tuned so that the model sits exactly at half filling without us needing to add and adjust a separate μ by hand.

Two limits of Eq. (2) are easy to solve outright. At $V = 0$ we just have the free tight-binding chain of Session 1: a single cosine band, filled up to the Fermi points, a metal with a unique, non-degenerate ground state. At $V \rightarrow \infty$ we can drop the hopping altogether, and the interaction alone is minimized by the classical alternating pattern $1, 0, 1, 0, \dots$: exactly one electron every other site. There are exactly *two* such states, related by shifting the pattern by one site, and they are degenerate: this is the charge-density wave, and its defining signature is that the ground state is twofold degenerate, in sharp contrast with the unique metallic ground state at $V = 0$.

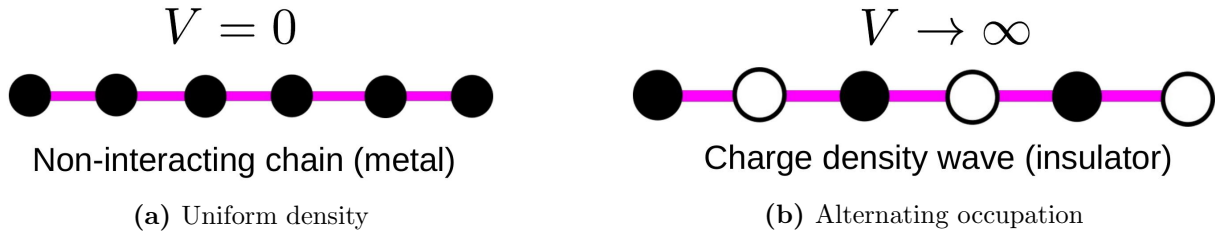


Figure 2: The two solvable limits of Eq. (2), with filled circles marking occupied sites and open circles empty ones. (a) At $V = 0$ the chain is a free tight-binding metal at half filling, with the same average density on every site and a unique ground state. (b) As $V \rightarrow \infty$ the hopping becomes irrelevant and the interaction alone selects the alternating $1, 0, 1, 0, \dots$ pattern. Shifting that pattern by one site produces a second state of identical energy, and this twofold degeneracy is the defining signature of the charge-density wave.

This is precisely what exact diagonalization lets us watch happen on a finite chain. Diagonalize Eq. (2) at fixed L and plot the low-lying eigenvalues, measured relative to the ground state, as a function of V . At small V we see a unique ground state followed by a small gap to a dense set of excited states: this gap is a finite-size artifact, roughly the level spacing $\sim t/L$ of the underlying free-fermion band, and it visibly shrinks as we go from $L = 12$ to $L = 14$, exactly as we would expect if it is destined to close in the thermodynamic limit and leave behind a continuum. As V grows, one particular excited state peels away from that continuum and drops steadily toward the ground state, becoming exactly degenerate with it as $V \rightarrow \infty$: this is the CDW partner state, and unlike the metallic gap, this near-degeneracy does *not* close by adding more excited states underneath it; instead a genuinely large gap opens above the (now twofold) ground state, growing asymptotically as $\Delta \sim V$. That linear growth has a simple interpretation: at large V , the cheapest way to excite the system is to introduce a single domain wall into the alternating pattern, a local defect where the charge order runs into itself, and each domain wall costs an energy of order V regardless of the size of the system, exactly the extensive, size-independent gap we see numerically.

It is worth comparing this directly with Session 1’s mean-field treatment of the very same model. There, the self-consistent gap also opened continuously as V increased and also grew linearly at large V , and here exact diagonalization confirms both of those features quantitatively. This is not an accident: the CDW breaks a *discrete* symmetry (translation by one site, a $\mathbb{Z}_2$ symmetry), and discrete symmetry breaking of this kind is allowed even in a one-dimensional quantum system at zero temperature. Keep this in mind, because in Section 4 we will meet a case where mean-field theory tries to break a *continuous* symmetry, and there the story changes completely.

✕ Computational companion

Sections “Many-body energies of an interacting fermionic model” and “Towards interaction-induced symmetry breaking.” Plot the low-lying many-body spectrum of the spinless chain as a function of V for two system sizes, and identify by eye where the first excited state collapses onto the ground state, this is the CDW transition. Then look at how the large- V gap scales with V and check the linear behavior we derived above.

3 Chemical potential, conserved quantum numbers, and the charging energy

So far we have kept the electron number fixed at half filling. Let us now add a chemical potential explicitly,

$$\mathcal{H}(\mu) = \mathcal{H}_{\text{CDW}} - \mu \sum_i n_i \equiv \mathcal{H}_{\text{CDW}} - \mu N, \quad (3)$$

where $N = \sum_i n_i$ is the total particle number operator. The key structural fact here is that $[\mathcal{H}_{\text{CDW}}, N] = 0$: hopping moves an electron from one site to another without changing how many there are in total, and the interaction only counts electrons, so it commutes with N trivially too. Since $\mathcal{H}_{\text{CDW}}$ and N commute, they share a common eigenbasis, which means adding the $-\mu N$ term does not change a single eigenvector of $\mathcal{H}_{\text{CDW}}$: it only shifts each eigenenergy by $-\mu N$, where N is that eigenstate’s own, fixed particle number. Concretely, this means every eigenvalue of $\mathcal{H}(\mu)$ traces out a straight line as a function of μ , with slope $-N$ set entirely by that state’s particle number relative to the ground state’s.

This has an immediate and useful consequence: because different excited states carry different particle numbers, they have different slopes, and as we sweep μ these lines cross. Right at a crossing, the identity of the *ground state* itself changes: below the crossing the lowest-energy state has N electrons, above it the lowest-energy state has $N \pm 1$. In other words, μ is a knob that directly controls the electron number of the ground state, exactly as a gate voltage does in a real quantum dot or nanostructure.

Let us make this quantitative. Define $E(N)$ as the lowest eigenvalue of $\mathcal{H}_{\text{CDW}}$ restricted to the N -particle sector. The state with N particles is the ground state of $\mathcal{H}(\mu)$ precisely when $E(N) - \mu N$ is lower than $E(N \pm 1) - \mu(N \pm 1)$, which rearranges into a window,

$$E(N) - E(N - 1) < \mu < E(N + 1) - E(N). \quad (4)$$

The width of this stability window is exactly the **charging energy**,

$$U_c(N) \equiv [E(N + 1) - E(N)] - [E(N) - E(N - 1)], \quad (5)$$

the discrete second derivative of the ground-state energy with respect to particle number. It measures how much harder it gets to add one more electron than it was to add the previous one, and it is exactly the quantity that produces the flat plateaus you see when you plot the ground-state occupation $\langle N \rangle$ against μ : on a plateau, μ can move freely without changing N , because it has not yet exceeded the addition energy in Eq. (4). As V grows, the interaction makes it costlier to double up charge on already-crowded sites, U_c grows with it, and the plateaus in the $\langle N \rangle$ -versus- μ curve widen accordingly, exactly what the notebook shows you when you compare a weakly and a strongly interacting chain.

✕ Computational companion

Sections “Energies of an interacting fermionic model as a function of the chemical potential,” “Quantum numbers in the interacting fermionic model,” “Level crossings in the interacting fermionic model,” and “Electronic density as a function of the chemical potential.” Track the particle number of each many-body eigenstate as μ is swept, watch the ground state switch particle number at level crossings, and confirm that the width of each occupation plateau grows with the interaction strength, exactly the charging-energy statement of Eq. (5).

4 The exact Hubbard magnetization: mean-field theory fails again

Now let us bring spin back and repeat this exercise for the Hubbard chain,

$$\mathcal{H}_{\text{Hub}} = -t \sum_{i,\sigma} \left(c_{i\sigma}^\dagger c_{i+1\sigma} + \text{h.c.} \right) + U \sum_i \left(n_{i\uparrow} - \frac{1}{2} \right) \left(n_{i\downarrow} - \frac{1}{2} \right), \quad (6)$$

which, just as in Eq. (2), differs from Session 1’s Hubbard Hamiltonian only by the same book-keeping shift, an automatic chemical potential that pins us at half filling.

Here is the punchline, and it is worth sitting with: if we diagonalize $\mathcal{H}_{\text{Hub}}$ exactly and compute the local magnetization $m_i \equiv \langle n_{i\uparrow} \rangle - \langle n_{i\downarrow} \rangle$ site by site, we find $m_i = 0$ *identically*, on every site, for every U , no matter how strong the interaction is. This flatly contradicts what Session 1’s mean-field theory told us to expect: there, the self-consistency loop converges, for any $U > 0$ on

a bipartite half-filled chain, to a finite staggered moment, $m_i = (-1)^i m$ with m growing toward saturation as $U \rightarrow \infty$. Exact diagonalization and mean-field theory are not just quantitatively different here, they disagree about whether the ground state is magnetic at all.

This is not a numerical fluke; it follows from an exact symmetry argument. The Hamiltonian Eq. (6) is invariant under any global spin rotation, so it commutes with the total spin operators, $[\mathcal{H}_{\text{Hub}}, \mathbf{S}^2] = [\mathcal{H}_{\text{Hub}}, S_z^{\text{tot}}] = 0$, and its eigenstates can be labeled by a definite total spin S . **Lieb's theorem** pins this down completely for the repulsive Hubbard model on any bipartite lattice at half filling: the ground state is unique and has total spin $S = \frac{1}{2}|N_A - N_B|$, where N_A, N_B are the sizes of the two sublattices. On our chain the sublattices are equal in size, so $S = 0$ exactly, a spin singlet, for every U and every chain length. A rotationally invariant, unique, total-spin-zero state simply cannot have a nonzero expectation value of S_i^z on any individual site, since a spin rotation would map it to an equally valid ground state with the opposite sign of m_i , and uniqueness forbids two different ground states. Mean-field theory, by construction, searches only over single Slater determinants, and a Slater determinant with staggered spin *explicitly* breaks spin-rotation symmetry from the very first guess: it can never represent a rotationally invariant, entangled singlet, exactly the same structural failure we met with the Hubbard dimer in Session 1.

Turning on an explicit symmetry-breaking field. We can watch this failure mode directly by interpolating between the exact model and the mean-field ansatz. Add an explicit staggered field,

$$\mathcal{H}(h) = \mathcal{H}_{\text{Hub}} + h \sum_i (-1)^i (n_{i\uparrow} - n_{i\downarrow}), \quad (7)$$

which is precisely the single-particle term that mean-field theory would have generated on its own had it found a self-consistent staggered solution (compare Session 1's $\mathcal{H}_{\text{MF}}$). At $h = 0$ we recover the exact result, $m = 0$. As h increases from zero, $m(h)$ grows *continuously* rather than jumping to a finite value: quantum fluctuations, encoded in the entangled singlet ground state, are actively fighting the ordering field, and only once h is large enough to overwhelm those fluctuations does $m(h)$ approach the mean-field saturation value. This is exactly the standard diagnostic for spontaneous symmetry breaking: the true order parameter is defined as $m \equiv \lim_{h \rightarrow 0^+} \lim_{L \rightarrow \infty} m(h, L)$, and the order of these two limits matters enormously. At any finite L , symmetry forces $m(h=0, L) = 0$ exactly, so we must send $L \rightarrow \infty$ *first*. Even having done that, for the one-dimensional Heisenberg antiferromagnet that this model reduces to at large U (Section 5), this limit still gives zero: one dimensional quantum antiferromagnets do not sustain true long-range magnetic order, only quasi-long-range, power-law spin correlations, which is exactly the gaplessness we will meet again in Section 6. This is the deep reason mean-field theory fails so badly here: it is not just missing quantitative corrections, it is predicting an ordered phase that, in one dimension, does not exist.

	CDW (Section 2)	Hubbard magnetism (this section)
Symmetry broken	discrete, $\mathbb{Z}_2$ (one-site translation)	continuous, $SU(2)$ (spin rotation)
Mean-field prediction	finite δn , growing continuously with V	finite staggered m , growing continuously with U
Exact 1D result	agrees qualitatively: order onset, gap $\sim V$	disagrees: $m = 0$ identically, unique spin singlet

Table 1: The two symmetry-breaking scenarios met so far, contrasted: which symmetry is broken, what mean-field theory predicts, and whether the exact one-dimensional result agrees.

Table 1 lines up the two symmetry-breaking scenarios met so far side by side: mean-field theory gets the discrete CDW transition qualitatively right, but gets the continuous Hubbard magnetization badly wrong.

✕ Computational companion

Sections “Magnetization of an interacting Hubbard model” and “Magnetization of a Hubbard model with stagger magnetization.” First confirm $m_i = 0$ at every site of the exact Hubbard ground state, however large U is. Then switch on the staggered field of Eq. (7) and watch m grow smoothly from zero, and check how the field strength needed to saturate m changes as you increase U .

5 Recovering the Heisenberg model: an exact worked example

Let us now directly verify, by exact diagonalization, the superexchange result $J \sim t^2/U$ that Session 1 derived by degenerate perturbation theory. Take the Hubbard dimer, $L = 2$, at half filling, and work in the sector with zero total S_z , spanned by the four states

$$|1\rangle = c_{0\uparrow}^\dagger c_{1\downarrow}^\dagger |\Omega\rangle, \quad |2\rangle = c_{0\downarrow}^\dagger c_{1\uparrow}^\dagger |\Omega\rangle, \quad |3\rangle = c_{0\uparrow}^\dagger c_{0\downarrow}^\dagger |\Omega\rangle, \quad |4\rangle = c_{1\uparrow}^\dagger c_{1\downarrow}^\dagger |\Omega\rangle, \quad (8)$$

the same singly- and doubly-occupied configurations we met in Session 1. Hopping connects a singly occupied state to a doubly occupied one but never $|1\rangle$ directly to $|2\rangle$, or $|3\rangle$ directly to $|4\rangle$, so in this basis $\mathcal{H}_{\text{Hub}}$ (up to the constant $U/2$ shift from Eq. (6): the singly-occupied states sit at $-U/2$ and the doubly-occupied ones at $+U/2$, so adding $U/2$ uniformly reproduces the diagonal below) is

$$\mathcal{H}_{\text{Hub}} \longrightarrow \begin{pmatrix} 0 & 0 & -t & t \\ 0 & 0 & t & -t \\ -t & t & U & 0 \\ t & -t & 0 & U \end{pmatrix} \quad \text{in the basis } \{|1\rangle, |2\rangle, |3\rangle, |4\rangle\}. \quad (9)$$

This block diagonalizes cleanly if we pass to symmetric and antisymmetric combinations, $|a\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |2\rangle)$ (the $S_z = 0$ member of the triplet), $|b\rangle = \frac{1}{\sqrt{2}}(|1\rangle - |2\rangle)$ (the singlet), and likewise $|c\rangle = \frac{1}{\sqrt{2}}(|3\rangle + |4\rangle)$, $|d\rangle = \frac{1}{\sqrt{2}}(|3\rangle - |4\rangle)$ for the doubly-occupied states. A short calculation shows $|a\rangle$ and $|c\rangle$ are already exact eigenstates,

$$\mathcal{H}_{\text{Hub}} |a\rangle = 0 \cdot |a\rangle, \quad \mathcal{H}_{\text{Hub}} |c\rangle = U |c\rangle, \quad (10)$$

while $|b\rangle$ and $|d\rangle$ mix into a genuine 2×2 block,

$$\mathcal{H}_{\text{Hub}}|_{\{b,d\}} = \begin{pmatrix} 0 & -2t \\ -2t & U \end{pmatrix} \Rightarrow E_{\pm} = \frac{U \pm \sqrt{U^2 + 16t^2}}{2}. \quad (11)$$

The lower eigenvalue, E_- , is the exact ground state, and comparing it against the exactly degenerate $S_z = \pm 1$ states $|\uparrow\uparrow\rangle, |\downarrow\downarrow\rangle$ (both sitting at energy 0, just like $|a\rangle$, since Pauli exclusion blocks all hopping for parallel spins), we see immediately that the full ground state *is* the total spin singlet ($S = 0$), sitting strictly below the $S = 1$ triplet at energy 0, exactly as Lieb's theorem guaranteed in Section 4. The exact singlet-triplet gap is therefore

$$\Delta_{s-t} = 0 - E_- = \frac{\sqrt{U^2 + 16t^2} - U}{2} \xrightarrow{U \gg t} \frac{4t^2}{U}, \quad (12)$$

which reproduces, from a completely independent, exact calculation, not just the t^2/U scaling but the exact same numerical prefactor that Session 1 obtained perturbatively for the effective coupling $J = 4t^2/U$ of $\mathcal{H}_{\text{eff}} = J \mathbf{S}_1 \cdot \mathbf{S}_2$: expanding Eq. (12) to leading order in t/U gives $\Delta_{s-t} \rightarrow 4t^2/U$, digit for digit. This is a genuinely satisfying check: two completely different techniques, one perturbative and spin-only, one exact and fermionic, land on exactly the same physics, not just the same scaling.

Excitations as magnons. We can probe the excited states further, exactly as in Session 2, by adding a Zeeman field $-h S_z^{\text{tot}}$. Since $[\mathcal{H}_{\text{Hub}}, S_z^{\text{tot}}] = 0$, this again leaves every eigenvector untouched and only shifts eigenenergies linearly in h , by an amount set by each state's own S_z^{tot} . The three states we called the triplet split apart linearly, into $S_z^{\text{tot}} = -1, 0, +1$ branches, while the singlet ground state, having $S_z^{\text{tot}} = 0$, does not move at all. This is precisely the field dependence of the Heisenberg dimer from Session 2, and the physical interpretation is the same either way: the lowest excitations above a singlet ground state are spin-1 objects, magnons, and their Zeeman splitting is a direct, numerically exact fingerprint of the underlying spin physics living inside the fermionic Hubbard model. The same mapping goes through, qualitatively, on longer chains: the exact fermionic spectrum at large U/t reproduces the exact spin spectrum of a Heisenberg chain with $J_{ij} \sim t_{ij}^2/U$, which is exactly what it means to say the two models are equivalent at strong coupling.

✕ Computational companion

Section “Connection between the Hubbard and Heisenberg models.” Diagonalize the Hubbard model at increasing U/t and compare its low-energy many-body spectrum directly against the Heisenberg spectrum from Session 2's notebook. Check that the states that map onto each other converge as U grows, and see how quickly the mapping breaks down as you push U back down toward t .

6 Spin-charge separation

We now have all the pieces to understand one of the most striking features of the one-dimensional Hubbard model: at strong coupling, its charge and spin degrees of freedom behave in completely different ways.

6.1 The many-body gap is not one thing

Equation (12) already told us the singlet-triplet gap of the dimer shrinks like $1/U$ at large U . If we track the same gap as a function of chain length L , this trend persists and even sharpens: at $U = 0$ the gap is a purely single-particle, finite-size level spacing, and it closes as a clean power law, $\Delta \sim 1/L$. Turning on U makes the gap *smaller* at fixed L , and increasing U further shrinks it more, which looks backwards at first: we expect the Hubbard model to become more insulating, not less gapped, as U grows. The resolution is that this particular gap is the **spin gap**: it is the energy to flip the total spin from $S = 0$ to $S = 1$, and Section 5 already told us this is governed by $J \sim t^2/U$, which indeed shrinks with U . In the thermodynamic limit this spin gap vanishes altogether, because the Heisenberg chain that the Hubbard model maps onto at strong coupling is itself gapless. None of this says anything yet about the **charge gap**, the energy to add or remove an electron, and that is a completely different quantity that we need to probe separately.

6.2 Nonlocal correlators: telling a metal from an insulator

A clean, ground-state-only way to distinguish a metal from an insulator is to look at how a static correlator decays with distance. Define the nonlocal particle correlator and the nonlocal spin correlator,

$$C_c(i, j) \equiv \langle c_{i\sigma}^\dagger c_{j\sigma} \rangle, \quad C_s(i, j) \equiv \langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle, \quad (13)$$

both evaluated in the many-body ground state. Physically, $C_s(i, j)$ measures how strongly the spin at site i is correlated with the spin at site j . Even at strong coupling, where the model reduces to the gapless Heisenberg chain, $C_s(r)$ decays as a power law rather than exponentially, precisely because the spin sector has no gap to suppress long-range propagation. What does grow with U is the *amplitude* of this correlator, not its decay rate: at $U = 0$ each site's spin is only weakly and loosely correlated with its neighbors (itinerant electrons barely notice each other's spin), while at large U every site carries an increasingly well formed, singly-occupied local moment, strongly antiferromagnetically correlated with its neighbors, approaching the ideal Heisenberg chain's correlations.

$C_c(i, j)$, by contrast, measures the amplitude to inject an electron at site i and later find it at site j : in a metal this decays only as a power law, $C_c(r) \sim \sin(k_F r)/r$, reflecting the fact that gapless charge excitations let the particle propagate arbitrarily far at essentially no energy cost. In an insulator, by contrast, any propagation has to borrow energy of order the charge gap Δ_c , and the correlator instead decays exponentially, $C_c(r) \sim e^{-r/\xi_c}$, with a correlation length ξ_c set by Δ_c (roughly $\xi_c \sim v/\Delta_c$ for some characteristic velocity v). This is exactly the signature we see in the Hubbard chain: at $U = 0$, $C_c(r)$ decays slowly, a power law, while at large U (say $U = 10t$) it collapses to zero within a couple of sites, an unmistakable exponential decay confirming the charge sector is gapped and insulating, exactly consistent with the linearly growing charge scale we will see again below. So interactions act in opposite directions on the two channels: they strongly suppress charge fluctuations while they strengthen spin correlations, even though the associated energy scale $J \sim t^2/U$ of those very same spin fluctuations is shrinking.

✕ Computational companion

Sections “Gap of an interacting Hubbard model,” “Non-local static correlators for the spin response,” and “Non-local static correlators for the particle-particle response.” Plot the singlet-triplet gap against L for a few values of U and confirm the $1/U$ trend derived above. Then plot $C_c(1, j)$ and $C_s(1, j)$ against j for weak and strong U , and check which one turns from power-law to exponential decay as U grows, that is your direct numerical evidence for a charge gap coexisting with a gapless spin sector.

6.3 Dynamical correlators and the two energy scales

Static correlators tell us about decay lengths; dynamical correlators tell us about the actual excitation energies involved, and are the many-body generalization of the single-particle density of states from Session 1. Define the dynamical particle and spin correlators via their Lehmann (spectral) representations,

$$S_c(\omega) = \sum_n \left| \langle n | c_{i\sigma}^\dagger | 0 \rangle \right|^2 \delta(\omega - (E_n - E_0)), \quad S_s(\omega) = \sum_n \left| \langle n | S_i^+ | 0 \rangle \right|^2 \delta(\omega - (E_n - E_0)), \quad (14)$$

where the sums run over all many-body eigenstates $|n\rangle$ of $\mathcal{H}_{\text{Hub}}$. $S_c(\omega)$ picks out the excitation energies reachable by adding a single electron, exactly the charge sector; $S_s(\omega)$ picks out excitation energies reachable by flipping a single spin, exactly the spin sector. At a large but fixed interaction, say $U = 60t$, the two look dramatically different: S_c has essentially no weight until a large threshold energy, and that threshold keeps drifting upward as U increases; S_s has weight starting at a much lower energy, and its overall lineshape closely tracks the Heisenberg-chain spin structure factor from Session 2, exactly what we expect from the strong-coupling mapping.

Scanning the peak position of each correlator as a function of U makes the contrast completely explicit: the leading charge peak moves to higher energy *linearly* in U (it costs $\sim U/2$ just to overcome the local repulsion when adding a second electron to an already singly-occupied site), while the leading spin peak moves to *lower* energy, decreasing $\sim 1/U$ as we derived in Section 5. Charge and spin excitations are governed by opposite trends in the very same parameter U , growing apart in energy without bound as U increases. This is what we mean by **spin-charge separation**: in the strongly correlated one-dimensional Hubbard chain, charge transport is exponentially suppressed, essentially frozen out in the thermodynamic limit, while spin transport remains gapless and, in this precise sense, easy, even though both live on the very same lattice of electrons.

✕ Computational companion

Sections “Dynamical particle-particle correlators in the Hubbard model,” “Dynamical spin-spin correlators in the Hubbard model,” and “Interaction dependence of the dynamical excitations in the Hubbard model.” Compute both dynamical correlators at a large, fixed U and compare their lineshapes, then track how the leading peak of each one moves as you scan U , confirming the linear-growth-versus- $1/U$ -decay contrast that defines spin-charge separation.

Summary

- Exact diagonalization directly in the fermionic Fock space costs 2^L (spinless) or 4^L (spinful) in Hilbert space dimension, exactly matching the 2^L cost of the equivalent spin problem

from Session 2, and it comes with no product-state restriction whatsoever.

- The spinless nearest-neighbor chain shows an exact charge-density-wave transition: a unique metallic ground state at $V = 0$ becomes a twofold-degenerate, gapped insulator as $V \rightarrow \infty$, with a large- V gap growing $\sim V$, in good qualitative and quantitative agreement with Session 1's mean-field treatment, since the broken symmetry here is only discrete.
- A conserved chemical potential term lets us control the ground-state particle number directly; the width of the resulting occupation plateaus is the charging energy, the discrete second derivative of the ground-state energy with respect to N .
- The exact Hubbard-model magnetization is *zero* at every site, for every U , in sharp and structural disagreement with Session 1's mean-field prediction of finite staggered order; Lieb's theorem guarantees a unique, rotationally invariant spin singlet ground state at half filling, something no single Slater determinant can ever represent. This is the clean, numerically exact confirmation of mean-field theory's failure flagged at the end of Session 1.
- An exact diagonalization of the Hubbard dimer reproduces the t^2/U superexchange scaling of Session 1 from first principles, and adding a Zeeman field shows the low-energy excitations are magnons, exactly matching the Heisenberg dimer of Session 2.
- At strong coupling the Hubbard chain exhibits spin-charge separation: the charge sector is gapped, with a gap growing linearly in U and exponentially decaying nonlocal correlators, while the spin sector stays gapless, with a spin gap shrinking as $1/U$ and power-law correlators, exactly the physics of the Heisenberg chain it maps onto.
- All of this was limited to $L \lesssim 15$ -ish sites by the $2^L/4^L$ wall. Session 5 returns to this same fermionic program, via the Jordan–Wigner mapping and tensor-network methods, and pushes it to system sizes where the thermodynamic limit becomes directly visible.